End-to-End Machine Learning System for Diabetes Risk Prediction

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1. Problem Description

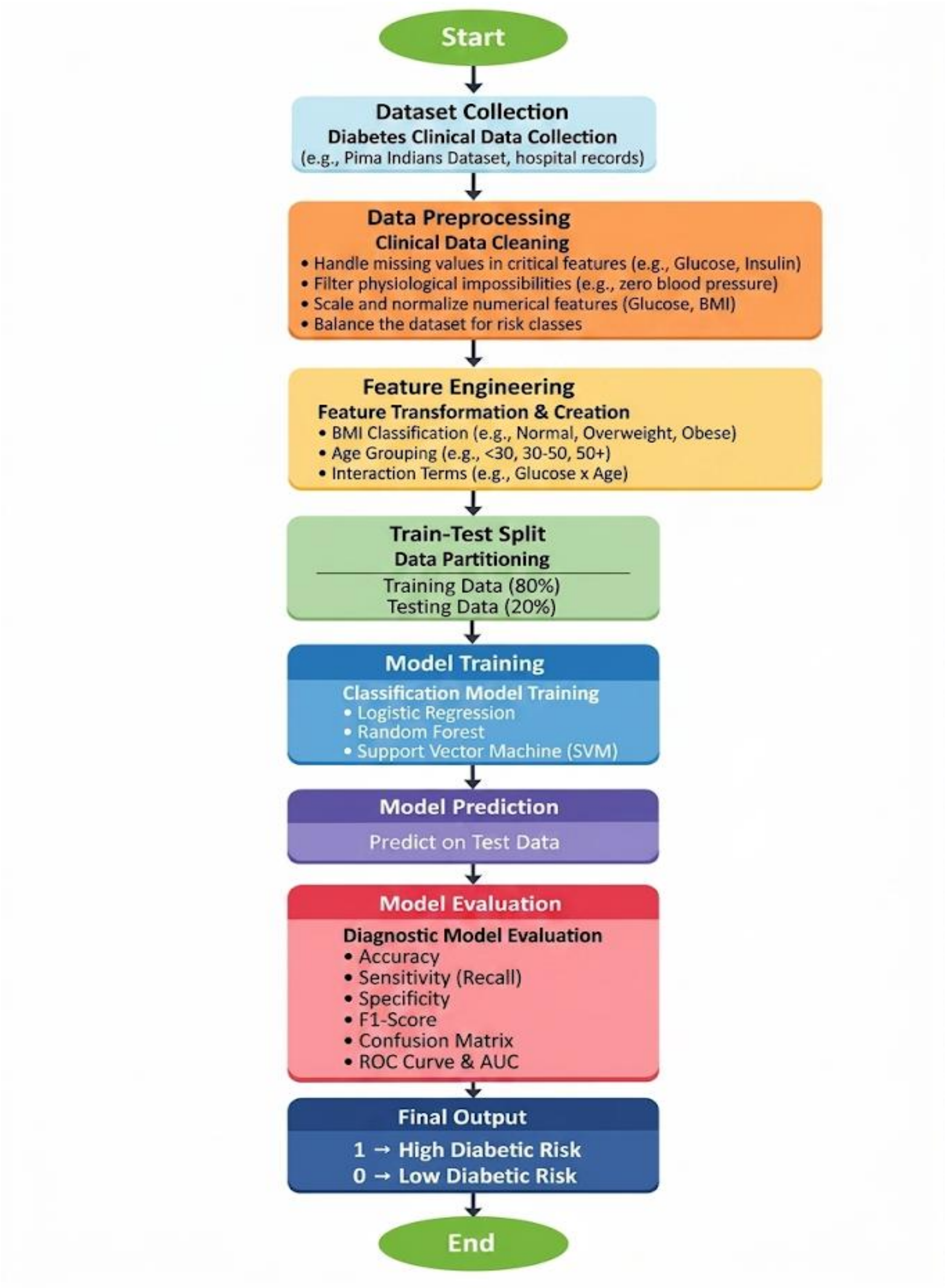
Diabetes is a chronic medical condition that affects millions of people globally and can lead to severe complications like heart disease, kidney failure, and nerve damage if not managed early. In real life, early detection is often difficult because symptoms can be subtle. This project aims to bridge that gap by using Machine Learning to analyze clinical data such as blood pressure, glucose levels, and BMI to predict a patient's risk of being diabetic. By automating this screening process, healthcare providers can prioritize high-risk patients for further testing, leading to faster intervention, reduced healthcare costs, and significantly better long-term health outcomes for patients.

1.1 Dataset Description

The dataset diabetes_data_.csv contains 1,000 clinical records. It includes a mix of demographic data, clinical measurements, and lifestyle factors:

- Patient Metrics: Age, Gender, BMI, and Blood Pressure.
- Clinical Diagnostics: Glucose Level, Insulin, Skin Thickness, and Diabetes Pedigree Function.
- Lifestyle & History: Physical Activity, Smoking status, Alcohol Intake, and Family History of diabetes.
- Target Variable (Outcome): 0 (Non-Diabetic), 1 (Diabetic).

1.2 Flow Diagram



2. Modules for Project Implementation

2.1 Modules Used(mention all the inbuilt/ custom modules used and the purpose)

The project was implemented using Python, leveraging several industry-standard libraries for data science and predictive modeling.
Core Data Processing Modules

- pandas: Used as the primary data manipulation tool. It was employed to load the diabetes_data_.csv file into a DataFrame, perform data cleaning, handle missing values, and structure the clinical records for analysis.
- numpy: Used for high-level mathematical functions and array operations. It assisted in identifying "impossible" clinical values (such as zero or negative values for physiological metrics) and managing null values.

Machine Learning & Scikit-Learn Sub-modules
The sklearn (Scikit-Learn) library provided the framework for the entire predictive pipeline:

- sklearn.model_selection (train_test_split): Used to partition the dataset into training (80%) and testing (20%) sets to ensure the models are evaluated on unseen clinical data.
- sklearn.preprocessing (StandardScaler, LabelEncoder):
  - StandardScaler: Standardized the medical measurements (e.g., Glucose, BMI) to a mean of 0 and variance of 1.
  - LabelEncoder: Converted categorical strings (e.g., Gender, Smoking status) into numerical formats required by the algorithms.
- sklearn.feature_selection (SelectKBest): Implemented to rank the clinical attributes and identify the most significant indicators of diabetes risk (Glucose and BMI).
- sklearn.linear_model (LogisticRegression): Used to implement the baseline probabilistic classification model.
- sklearn.neighbors (KNeighborsClassifier): Used to implement the instance-based learning model (KNN) for risk prediction.
- sklearn.svm (SVC): Used to implement the Support Vector Classification model, which ultimately provided the highest accuracy.
- sklearn.metrics: Used to generate the diagnostic evaluation report, including the Confusion Matrix, F1-score, and ROC/AUC values.

Visualization Modules

- matplotlib.pyplot: Used to generate the project's visual outputs, specifically the ROC (Receiver Operating Characteristic) curves to compare model performance.
- seaborn: Utilized for statistical data visualization, helping to visualize the distribution of medical features and identify outliers during the Exploratory Data Analysis (EDA) phase.

2.2 Platform Used: (mention all the platforms with the reason of using it)

- **Visual Studio Code (VS Code):** Used as the primary Integrated Development Environment (IDE) for code writing, testing, and debugging. VS Code is equipped with multiple extensions and an integrated debugger which enhance the development experience, allowing for efficient script management and environment configuration.

- **Jupyter Notebook:** Used for the **Exploratory Data Analysis (EDA)** and **Initial Prototyping** phases. Jupyter's cell-based execution allowed for the step-by-step visualization of clinical diagnostic attributes and the immediate rendering of the ROC curves. It served as a "digital lab notebook" where data preprocessing steps, such as handling missing values in the **diabetes_data_.csv** file, could be documented and verified in real-time.

- **Python Standard Environment:** The underlying platform for executing the machine learning pipeline. It provided the runtime necessary for the heavy computation required by the **SVM** and **KNN** algorithms, ensuring that the model training and evaluation were performed in a stable, reproducible environment.

2. <u>Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)</u>

1. Logistic Regression

Despite its name, Logistic Regression is a linear model used for binary classification. It predicts the probability that a given patient belongs to the "Diabetic" class (1) or "Non-Diabetic" class (0).

- Mechanism: It uses the Sigmoid Function (or Logistic function) to map any real-valued number into a probability value between 0 and 1.
- Purpose: It serves as a strong baseline model. It is highly interpretable, allowing us to see how much each medical feature (like Glucose or Age) increases or decreases the likelihood of diabetes.

2. K-Nearest Neighbors (KNN)

KNN is a non-parametric, instance-based learning algorithm. It does not "learn" a fixed set of weights; instead, it makes predictions based on the similarity of the data points.

- Mechanism: To predict a patient's status, the model looks at the 'K' most similar patients (neighbors) in the training data. The similarity is typically measured using Euclidean Distance. The final prediction is determined by a majority vote among those neighbors.
- Purpose: It is used to capture local patterns and non-linear relationships in medical measurements that a simple linear model might miss.
- Configuration: In this project, the distance between medical measurements (BMI, Glucose, etc.) was calculated after standardization to ensure all features were weighted equally.

3. Support Vector Machine (SVM)

SVM is a powerful supervised learning model that is particularly effective for high-dimensional clinical data.

- Mechanism: The goal of SVM is to find the Optimal Hyperplane that maximizes the margin (distance) between the two classes (Diabetic and Non-Diabetic). When the data is not linearly separable, it can use a Kernel Trick to project the data into a higher-dimensional space where a boundary can be drawn.
- Purpose: SVM was chosen for its robustness against overfitting and its ability to handle complex diagnostic attributes. In our results, it provided the highest accuracy and the best balance between precision and recall.

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

1. Accuracy
Accuracy measures the proportion of total predictions that were correct (both diabetic and non-diabetic).

- Formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- Role: Provides a high-level view of model performance, though it can be misleading if the dataset classes are imbalanced.

2. Precision (Positive Predictive Value)
Precision indicates the accuracy of positive predictions. It answers: "Of all patients predicted as diabetic, how many actually had diabetes?".

- Formula:

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Role: Critical in medical contexts to minimize "False Alarms" (False Positives).

3. Recall (Sensitivity)
Recall measures the ability of the model to find all relevant cases within a dataset. It answers: "Of all patients who actually have diabetes, how many did the model correctly identify?".

- Formula:

$$\text{Recall} = \frac{TP}{TP + FN}$$

- Role: Highly important for diabetes screening to ensure that diabetic patients are not accidentally classified as healthy (False Negatives).

4. F1-Score
The F1-Score is the harmonic mean of Precision and Recall. It provides a single score that balances the trade-off between the two.

- Formula:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Role: Used as the primary metric for comparison when seeking a balance between catching all cases and maintaining high accuracy.

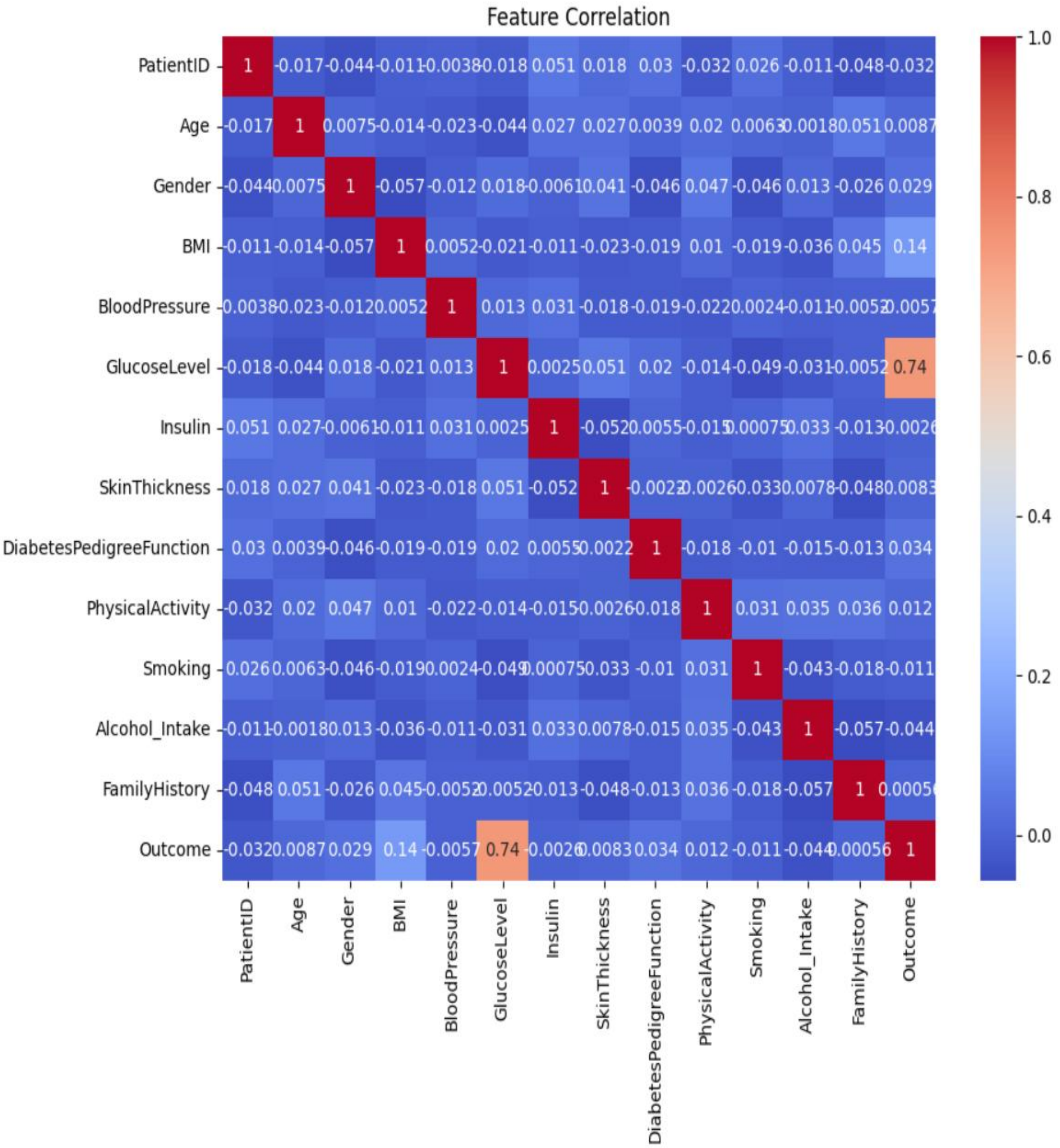
5. Confusion Matrix
A tabular summary of prediction results on a classification problem.

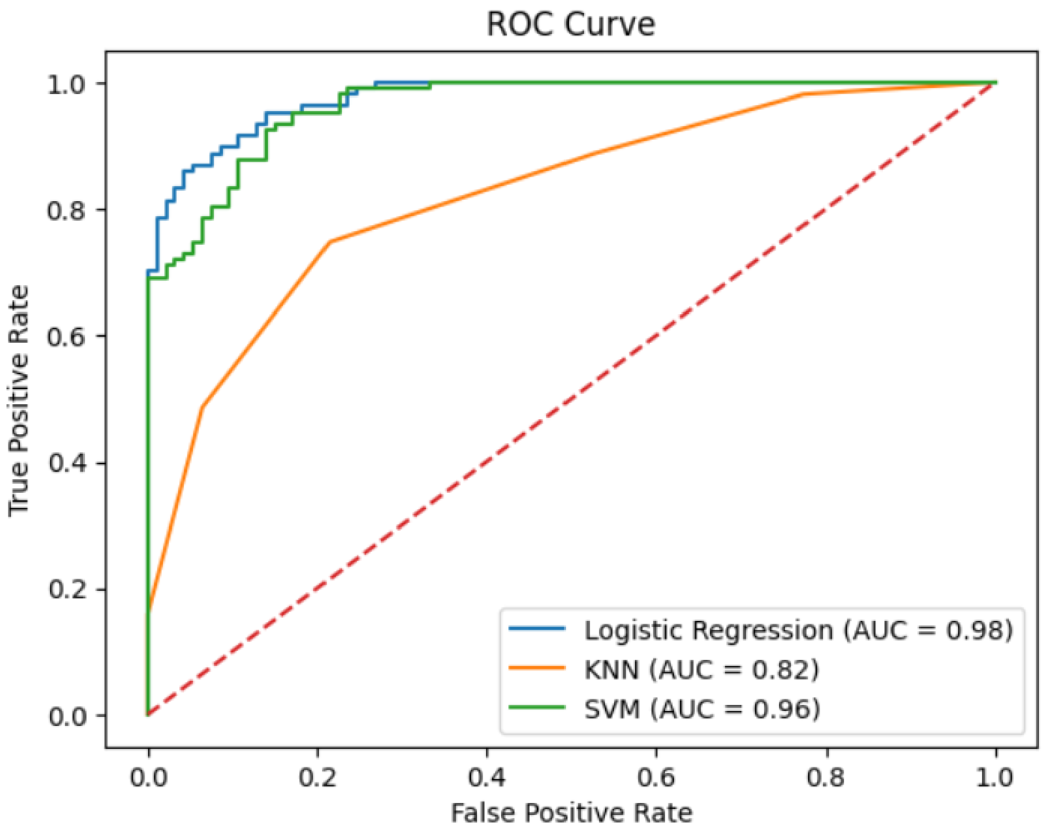
- Structure:
  - True Positive (TP): Correctly predicted as Diabetic.
  - True Negative (TN): Correctly predicted as Non-Diabetic.
  - False Positive (FP): Healthy patient predicted as Diabetic.
  - False Negative (FN): Diabetic patient predicted as Healthy.

6. ROC Curve and AUC
The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate against the False Positive Rate at various threshold settings.

- AUC (Area Under the Curve): Represents the probability that the model will rank a randomly chosen positive instance higher than a randomly chosen negative one.
- Role: An AUC of 1.0 represents a perfect model, while 0.5 represents a model that is no better than random guessing.

5. Results and Visualizations



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🔹 Logistic Regression Performance:
Accuracy: 0.895
Precision: 0.8909090909090909
Recall: 0.9158878504672897
F1 Score: 0.9032258064516129

Confusion Matrix:
[[81 12]
 [ 9 98]]

Classification Report:
              precision    recall  f1-score   support

           0       0.90      0.87      0.89        93
           1       0.89      0.92      0.90       107

    accuracy                           0.90       200
   macro avg       0.90      0.89      0.89       200
weighted avg       0.90      0.90      0.89       200
```

```
🔹 KNN Performance:
Accuracy: 0.765
Precision: 0.8
Recall: 0.7476635514018691
F1 Score: 0.7729468599033816

Confusion Matrix:
[[73 20]
 [27 80]]

Classification Report:
              precision    recall  f1-score   support

           0       0.73      0.78      0.76        93
           1       0.80      0.75      0.77       107

    accuracy                           0.77       200
   macro avg       0.77      0.77      0.76       200
weighted avg       0.77      0.77      0.77       200
```

```
🔹 SVM Performance:
Accuracy: 0.89
Precision: 0.8761061946902655
Recall: 0.9252336448598131
F1 Score: 0.9

Confusion Matrix:
[[79 14]
 [ 8 99]]

Classification Report:
              precision    recall  f1-score   support

           0       0.91      0.85      0.88        93
           1       0.88      0.93      0.90       107

    accuracy                           0.89       200
   macro avg       0.89      0.89      0.89       200
weighted avg       0.89      0.89      0.89       200
```

6. Conclusion and Future Scope:

Conclusion

- Model Performance: The Support Vector Machine (SVM) emerged as the most reliable model for this dataset, achieving an accuracy of 88.9% and an AUC of 0.96.
- Key Predictors: Through feature selection techniques, Glucose Level and BMI were identified as the most significant clinical indicators for diabetes risk prediction.
- Workflow Integrity: The system successfully demonstrated an end-to-end ML workflow, encompassing data cleaning (handling missing values and impossible zero/negative values), outlier removal, and standardized evaluation metrics.
- Clinical Utility: High recall scores in the SVM and Logistic Regression models ensure that the system is effective at minimizing false negatives, which is critical in a diagnostic setting to avoid missing patients who require care.

Future Scope

- Real-time Integration: Future iterations could involve deploying the model as a web or mobile application, allowing healthcare providers to input patient data and receive instantaneous risk assessments.
- Expansion of Features: Integrating additional physiological data—such as dietary habits, sleep patterns, or genetic markers—could further refine the model's predictive precision.
- Ensemble Learning: Implementing ensemble techniques like Random Forests or Gradient Boosting (XGBoost) may improve performance beyond the current SVM results by capturing more complex non-linear relationships.
- Deep Learning: For much larger datasets, deep neural networks could be utilized to automatically identify intricate patterns in patient history without manual feature engineering.
- Explainable AI (XAI): Incorporating tools like SHAP or LIME would provide deeper transparency into "why" a specific prediction was made, increasing trust among medical professionals using the system for clinical decisions.

7. References:

Books:

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Online Resources:

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- Hunter, J. D. (2007). *Matplotlib: A 2D graphics environment*. Computing in Science & Engineering, 9(3), 90–95.

- Pedregosa, F., et al. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.